

Representation Atlas and New Analytic Bridges for the Fabius Function, Rvachev’s Up-Function, and Their Fourier Images

Research report prepared by OpenAI GPT-5.6 Pro
Source audit: Vladimir Reshetnikov’s **ProveIt** repository

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Abstract

This report gives a source-audited atlas of integral, series, product, convolution, probabilistic, spectral, and transform representations of the bounded Fabius function F , Rvachev’s compactly supported even function up , the inverse Fabius function $Q = F^{-1}$, and the entire Fourier image $\Phi(z) = \widehat{up}(z)$. The baseline material is organized from the current formal exposition and the living non-formalized frontier documents in the **ProveIt** repository. Previously present results – including the sinc and cosine products, the arithmetic canonical product, Fredholm determinants, half-integer spectral series, q -binomial moment formulae, Thue–Morse splines, the generalized-gamma-convolution dual, and the logistic dual – are explicitly quarantined from novelty claims.

Five new-to-the-audited-corpus theorem packages are developed. First, a meromorphic *dyadic gamma function*

$$\Gamma_{dy}(z) = \prod_{h \geq 0} \Gamma(1 + 2^{-h}z)$$

factors the Rvachev transform as $\Phi(z) = (\Gamma_{dy}(z)\Gamma_{dy}(-z))^{-1}$, with functional equations, canonical products, digamma and heat-trace integrals, and a geometric- q extension. Second, the existing dyadic logistic series is identified in law with an explicit Gaussian variance mixture driven by the existing discrete Thorin random variable; this yields a convolution semigroup, a closed Levy density, complete monotonicity of the density as a function of x^2 , and dimension-free radial positive definiteness. Third, Jensen’s circular mean of $\log |\Phi|$ is evaluated exactly by binary digit sums and a finite factorial product. Fourth, the unbounded zero multiplicities prove that Φ is not D-finite, and neither are the dyadic gamma factors. Fifth, distributional and compact Fourier formulae for F are completed by an odd-harmonic sine series, while Q receives a new nonlinear Fourier transform and sine-series representation in terms of oscillatory integrals of F .

A fully commented Python program independently checks the principal identities, produces four figures, and records truncation diagnostics. “Proved here” means proved in this report and not found by targeted collision searches in the audited repository; it is not a claim of priority over all external literature.

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1 Scope, source audit, and status discipline

1.1 Repository boundary

The requested source boundary was the recursive collection of `*.tex` files below

<https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs>.

The primary formal exposition was pinned to repository commit `0770442945d65bac5530c4f214c6c52c4cb9fbd`, dated 27 August 2026. During the audit the branch changed: in particular, a support-free quantile substitution theorem entered the primary exposition. Consequently, quantile pushforward formulae that looked new against an earlier snapshot are classified here as repository consequences, not new results.

The recursive audit included the current formal exposition, the consolidated frontier notebook, the glossary and Lean walkthrough, the active frontier directories, the embedded literature sources, and the archive provenance map. The active frontier root contained seventeen named draft packages at the time of the audit. One of those packages records a snapshot with sixty-seven distinct TeX paths. Historical byte-identical copies were not counted as independent mathematical sources: the archive provenance file was used to map them to their living successors, after which the successor source was audited at formula level. This avoids manufacturing “novelty” merely by overlooking a renamed or moved draft.

1.2 Collision quarantine

The following structures were already present and are therefore baseline in this report:

- the infinite sinc product, weighted cosine product, and integer-zero canonical product;
- the arithmetic multiplicities $a_n = 1 + v_2(n)$, their Dirichlet series, counting law $\sum_{n \leq N} a_n = 2N - s_2(N)$, lobe signs, heat traces, spectral zeta functions, and Fredholm determinant;
- half-integer cosine expansions, spectral quadrature, sampling and alias formulae;
- moments and cumulants through Bell and Bernoulli numbers, q -binomial and multiple-zeta formulae, Appell sequences, Barnes-type prefixes, and Thue–Morse finite blocks;
- the discrete-Thorin generalized gamma convolution $\mathbb{E}e^{-wV_\tau} = \mathcal{A}(w)^{-\tau}$ and its complete-Bernstein/Stieltjes consequences;
- the dyadic logistic series whose characteristic function is the reciprocal moment generating function of the compact Rvachev law;
- exact negative-Laplace decompositions, periodic lower-Lambert endpoint asymptotics, inverse asymptotics, and the now-formal quantile substitution theorem.

Targeted searches did not find the phrases or exact structures “dyadic gamma”, “Gaussian variance mixture”, “Jensen circular mean”, “D-finite”, the principal-value transform of the global Fabius CDF, or the inverse sine coefficients derived in [Section 14](#). These results are marked [\[proved here\]](#). Formulae obtained by combining known repository theorems without a substantively new bridge are marked [\[derived synthesis\]](#).

1.3 Status legend and limits of the novelty statement

Status	Meaning
[repository]	Present in the formal exposition or an audited frontier document.
[derived synthesis]	A direct synthesis or reformulation of repository results; no novelty claim.
[proved here]	Proved in this report and not located by targeted collision searches in the audited corpus. This is a corpus-relative statement, not universal priority.
[conjecture]	A plausible but unproved statement, separated from theorems.

2 Normalization and the master probability diagram

2.1 The compact law and its CDF

Let $(V_j)_{j \geq 1}$ be independent uniform random variables on $[0, 1]$, let $U_j = 2V_j - 1$ be uniform on $[-1, 1]$, and define

$$X = \sum_{j \geq 1} 2^{-j} V_j, \quad Y = 2X - 1 = \sum_{j \geq 1} 2^{-j} U_j. \quad (1)$$

Then $X \in [0, 1]$, $Y \in [-1, 1]$, $X \stackrel{d}{=} 1 - X$, and $Y \stackrel{d}{=} -Y$. The bounded Fabius function and Rvachev density are

$$F(x) = \mathbb{P}(X \leq x), \quad \mathbb{P}(Y \in dy) = \text{up}(y) dy. \quad (2)$$

The normalizations used below are

$$\text{up}(y) = \begin{cases} F(y+1), & -1 \leq y \leq 0, \\ F(1-y) = 1 - F(y), & 0 \leq y \leq 1, \\ 0, & |y| > 1, \end{cases} \quad (3)$$

and

$$F'(x) = 2 \text{up}(2x - 1) \quad (0 < x < 1). \quad (4)$$

Thus up is even, nonnegative, C^∞ , supported on $[-1, 1]$, and has integral one. The density of X is $f_X(x) = 2 \text{up}(2x - 1)$ on $[0, 1]$.

2.2 Fixed-point integral equations

The random affine fixed point

$$Y \stackrel{d}{=} \frac{U + Y'}{2}, \quad (5)$$

where U is uniform on $[-1, 1]$ and Y' is an independent copy of Y , gives the exact refinement equation

$$\boxed{\text{up}(x) = \int_{2x-1}^{2x+1} \text{up}(t) dt.} \quad (6)$$

Differentiation gives

$$\text{up}'(x) = 2 \text{up}(2x + 1) - 2 \text{up}(2x - 1). \quad (7)$$

The bounded Fabius function is the fixed point of

$$F(x) = \begin{cases} \int_0^{2x} F(t) dt, & 0 \leq x \leq \frac{1}{2}, \\ 1 - \int_0^{2-2x} F(t) dt, & \frac{1}{2} \leq x \leq 1, \end{cases} \quad (8)$$

with constant continuation outside $[0, 1]$. Hence

$$F'(x) = \begin{cases} 2F(2x), & 0 < x < \frac{1}{2}, \\ 2F(2-2x), & \frac{1}{2} < x < 1. \end{cases} \quad (9)$$

These integral equations are often the most efficient starting point for exact quadrature, monotonicity, and contraction arguments.

2.3 The signed global extension and derivative tower

The repository's signed global extension $\mathcal{F}$ satisfies

$$\mathcal{F}'(x) = 2\mathcal{F}(2x), \quad (10)$$

so iteration gives the all-order identity

$$\boxed{\mathcal{F}^{(m)}(x) = 2^{\binom{m+1}{2}} \mathcal{F}(2^m x).} \quad (11)$$

After the appropriate unit translation, this yields

$$\text{up}^{(m)}(x) = 2^{\binom{m+1}{2}} \mathcal{F}(2^m(x+1)). \quad (12)$$

Thus every derivative representation of $\mathcal{F}$ immediately becomes one for up, and vice versa. It is important not to apply (10) directly to the bounded CDF outside its valid normalization.

3 Infinite convolutions, finite splines, and head–tail representations

3.1 Box convolution

The density of $2^{-j}U_j$ is $2^{j-1}\mathbf{1}_{[-2^{-j}, 2^{-j}]}$. Therefore

$$\boxed{\text{up} = \bigast_{j=1}^{\infty} \left(2^{j-1} \mathbf{1}_{[-2^{-j}, 2^{-j}]} \right).} \quad (13)$$

This identity may be read in probability, in L^1 , or through uniform convergence of successive compact splines. It directly explains positivity, compact support, smoothness, and the sinc product of Section 4.

Let

$$Y_N = \sum_{j=1}^N 2^{-j} U_j, \quad u_N = \bigast_{j=1}^N \left(2^{j-1} \mathbf{1}_{[-2^{-j}, 2^{-j}]} \right). \quad (14)$$

Then u_N is a piecewise polynomial of degree $N-1$ and

$$Y = Y_N + 2^{-N} Y', \quad \text{up} = u_N * \left(2^N \text{up}(2^N \cdot) \right). \quad (15)$$

Equivalently,

$$\text{up}(x) = \mathbb{E} \left[2^N \text{up}(2^N(x - Y_N)) \right]. \quad (16)$$

For the CDF,

$$F(x) = \mathbb{E} \left[F \left(2^N x - \sum_{j=1}^N 2^{N-j} V_j \right) \right], \quad (17)$$

where the inner F is the bounded, clamped CDF. Formula (17) is an exact N -dimensional cube integral and a useful recursive numerical scheme.

3.2 General truncated-power spline

For positive widths $a_1, \dots, a_N$, the density of $\sum_{j=1}^N a_j U_j$ is

$$b_N(x) = \frac{1}{(N-1)! 2^N \prod_{j=1}^N a_j} \sum_{\varepsilon \in \{\pm 1\}^N} \left(\prod_{j=1}^N \varepsilon_j \right) \left(x + \sum_{j=1}^N \varepsilon_j a_j \right)_+^{N-1}. \quad (18)$$

Its CDF is obtained by replacing $(N-1)!$ and the exponent $N-1$ by $N!$ and N . For $a_j = 2^{-j}$, binary indexing turns (18) into the exact Thue–Morse spline

$$u_N(x) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{k=0}^{2^N-1} (-1)^{s_2(k)} \left(x + \frac{2^N - 1 - 2k}{2^N} \right)_+^{N-1}. \quad (19)$$

The corresponding finite CDF is

$$U_N(x) = \frac{2^{N(N-1)/2}}{N!} \sum_{k=0}^{2^N-1} (-1)^{s_2(k)} \left(x + \frac{2^N - 1 - 2k}{2^N} \right)_+^N. \quad (20)$$

Prouhet cancellation makes the apparent polynomial tails cancel exactly. These formulae simultaneously represent the finite probability law, a compact spline, and a signed Thue–Morse power sum.

3.3 Hypercube and section representations

For bounded measurable g ,

$$\int_{-1}^1 g(y) \text{up}(y) dy = \int_{[-1,1]^N} g \left(\sum_{j \geq 1} 2^{-j} u_j \right) \bigotimes_{j \geq 1} \frac{du_j}{2}. \quad (21)$$

At finite depth this gives a coarea interpretation of u_N as the weighted $(N-1)$ -dimensional volume of the cube section $\sum_{j=1}^N 2^{-j} u_j = x$. The infinite-dimensional expression is often more useful than a closed formula: it converts integrals against up into product quadrature and supports concentration and coupling arguments.

4 Fourier-product atlas and spectral representations

4.1 Conventions and the three canonical products

We use

$$\widehat{g}(z) = \int_{\mathbb{R}} g(x) e^{-2\pi i x z} dx, \quad \text{sinc } w = \frac{\sin w}{w}, \quad (22)$$

with the removable value at zero. Put

$$\Phi(z) = \widehat{\text{up}}(z). \quad (23)$$

The repository establishes the entire products

$$\Phi(z) = \prod_{h=0}^{\infty} \text{sinc} \left(\frac{\pi z}{2^h} \right), \quad (24)$$

$$= \prod_{m=1}^{\infty} \left(\cos \frac{\pi z}{2^m} \right)^m, \quad (25)$$

$$= \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2} \right)^{a_n}, \quad a_n = 1 + v_2(n). \quad (26)$$

They satisfy the Mahler-type renormalization

$$\Phi(z) = \text{sinc}(\pi z) \Phi(z/2). \quad (27)$$

The zero at every nonzero integer n has exact order $a_{|n|}$.

4.2 Logarithmic Taylor series and partial fractions

For $|z| < 1$,

$$\log \Phi(z) = - \sum_{r \geq 1} \frac{\zeta(2r)}{r(1 - 4^{-r})} z^{2r}. \quad (28)$$

Logarithmic differentiation of the canonical product gives, away from the integer zeros,

$$\frac{\Phi'(z)}{\Phi(z)} = -2z \sum_{n \geq 1} \frac{a_n}{n^2 - z^2}. \quad (29)$$

The arithmetic spectrum has Dirichlet generating function

$$\sum_{n \geq 1} \frac{a_n}{n^s} = \frac{\zeta(s)}{1 - 2^{-s}}, \quad \Re s > 1, \quad (30)$$

and exact counting function

$$A(N) = \sum_{n \leq N} a_n = \sum_{h \geq 0} \left\lfloor N/2^h \right\rfloor = 2N - s_2(N). \quad (31)$$

Thus the zero divisor simultaneously encodes the Riemann zeta function, dyadic valuation, and binary digit sum.

4.3 Linear and quadratic heat traces

Two Lambert/heat kernels occur naturally:

$$L_1(t) = \sum_{n \geq 1} a_n e^{-nt} = \sum_{h \geq 0} \frac{1}{e^{2^h t} - 1}, \quad (32)$$

$$K(t) = \sum_{n \geq 1} a_n e^{-n^2 t} = \sum_{h \geq 0} \sum_{m \geq 1} e^{-4^h m^2 t}. \quad (33)$$

Their Mellin transforms are

$$\int_0^\infty L_1(t) t^{s-1} dt = \Gamma(s) \frac{\zeta(s)}{1 - 2^{-s}}, \quad \Re s > 1, \quad (34)$$

$$\int_0^\infty K(t) t^{s-1} dt = \Gamma(s) \frac{\zeta(2s)}{1 - 4^{-s}}, \quad \Re s > \frac{1}{2}. \quad (35)$$

The consolidated frontier work develops theta completion and complex-dimension oscillations for K . The linear kernel L_1 will reappear in the dyadic-gamma integral and in the Levy density of the Gaussian-mixture dual.

4.4 Frullani–Lambert integral for the Fourier product

Summing the elementary Frullani identity over the zero divisor gives the mixed integral/product representation

$$\log \Phi(z) = -2 \int_0^\infty (\cosh(zt) - 1) L_1(t) \frac{dt}{t}, \quad |\Re z| < 1. \quad (36)$$

In particular, for real y ,

$$\log \Phi(iy) = 2 \int_0^\infty (1 - \cos(yt)) L_1(t) \frac{dt}{t}. \quad (37)$$

This writes the Wick-rotated product as the exponential of a positive Lambert-kernel energy. It is also the Levy–Khinchine exponent that underlies [Section 10](#).

4.5 Fredholm determinant and spectral zeta

Let D be the positive diagonal operator whose eigenvalue n occurs with multiplicity a_n . Since D^{-2} is trace class,

$$\Phi(z) = \det(I - z^2 D^{-2}). \quad (38)$$

The associated spectral zeta function is

$$\zeta_D(s) = \text{tr}(D^{-s}) = \frac{\zeta(s)}{1 - 2^{-s}}, \quad \Re s > 1. \quad (39)$$

The determinant, zero multiplicities, heat trace, and arithmetic Dirichlet series are thus four views of the same spectrum.

5 Fourier inversion, sampling, and interval series

5.1 Whole-line inversion

Because $\text{up} \in C_c^\infty(\mathbb{R})$, its transform is rapidly decreasing on the real axis and

$$\text{up}(x) = \int_{\mathbb{R}} \Phi(\xi) e^{2\pi i x \xi} d\xi, \quad (40)$$

$$= 2 \int_0^\infty \Phi(\xi) \cos(2\pi x \xi) d\xi. \quad (41)$$

Every derivative may be obtained under the integral:

$$\text{up}^{(m)}(x) = \int_{\mathbb{R}} (2\pi i \xi)^m \Phi(\xi) e^{2\pi i x \xi} d\xi. \quad (42)$$

Combining this with (24), (26), or the dyadic-gamma factorization below gives distinct integral–product representations of the same bump.

5.2 Poisson partition and half-integer spectrum

The integer zeros imply the partition of unity

$$\sum_{k \in \mathbb{Z}} \text{up}(x - k) = 1. \quad (43)$$

On $[-1, 1]$, the half-integer samples give the rapidly convergent cosine expansion

$$\boxed{\text{up}(t) = \frac{1}{2} + \sum_{k \geq 0} \Phi\left(k + \frac{1}{2}\right) \cos((2k + 1)\pi t).} \quad (44)$$

The active half-integer spectral report develops weighted Prouhet identities, exact alias transforms, and spectral quadrature from this series.

5.3 Paley–Wiener sampling of the Fourier image

Since up is supported on $[-1, 1]$, Φ belongs to the Paley–Wiener space of bandwidth one. Shannon sampling at spacing $1/2$ gives

$$\Phi(z) = \sum_{k \in \mathbb{Z}} \Phi(k/2) \text{sinc}(\pi(2z - k)). \quad (45)$$

The rapid decay of the samples makes the series locally uniformly and absolutely convergent. Since $\Phi(m) = 0$ for every nonzero integer m ,

$$\Phi(z) = \text{sinc}(2\pi z) + \sum_{\substack{k \in \mathbb{Z} \\ k \text{ odd}}} \Phi(k/2) \text{sinc}(\pi(2z - k)). \quad (46)$$

Thus the entire transform is reconstructed from its half-integer arithmetic ray.

5.4 Fourier–Legendre representation

The compact interval also supports the uniformly convergent Legendre expansion

$$\text{up}(x) = \sum_{m \geq 0} \frac{2m+1}{2} \left(\int_{-1}^1 \text{up}(t) P_m(t) dt \right) P_m(x). \quad (47)$$

Evenness removes the odd indices. The coefficient can be expressed either as a moment combination or by inserting (41) and using the Fourier–Bessel transform of P_m . This is a useful bridge between exact moments and orthogonal approximation.

6 Laplace, Mellin, moment, and q -series representations

6.1 Moment-generating and Laplace products

The random series gives the entire moment-generating functions

$$M_Y(t) = \mathbb{E}e^{tY} = \Phi\left(\frac{it}{2\pi}\right) = \prod_{j \geq 1} \frac{\sinh(t/2^j)}{t/2^j}, \quad (48)$$

$$M_X(t) = \mathbb{E}e^{tX} = \prod_{j \geq 1} \frac{e^{t/2^j} - 1}{t/2^j}, \quad (49)$$

$$L_X(s) = \mathbb{E}e^{-sX} = \prod_{j \geq 1} \frac{1 - e^{-s/2^j}}{s/2^j}. \quad (50)$$

For $c > 0$, the CDF has the Bromwich representation

$$F(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{L_X(s)}{s} e^{sx} ds. \quad (51)$$

This contour formula is the natural entrance to the lower-Lambert saddle theory.

6.2 Mellin moments, including negative orders

For $\Re \alpha > 0$,

$$\mathbb{E}X^{-\alpha} = \frac{1}{\Gamma(\alpha)} \int_0^\infty s^{\alpha-1} L_X(s) ds. \quad (52)$$

The endpoint flatness makes every negative moment finite. For $0 < \Re \alpha < 1$,

$$\mathbb{E}X^\alpha = \frac{\alpha}{\Gamma(1-\alpha)} \int_0^\infty (1 - L_X(s)) s^{-\alpha-1} ds. \quad (53)$$

For integer $m \geq 1$, integration by parts gives the survival representation

$$\mathbb{E}X^m = m \int_0^1 x^{m-1} (1 - F(x)) dx = m \int_0^1 x^{m-1} \text{up}(x) dx. \quad (54)$$

6.3 Cumulants, Bernoulli numbers, and Bell polynomials

From (28), the nonzero cumulants of Y are

$$\kappa_{2r}(Y) = \frac{B_{2r}}{2r(1-4^{-r})} = (-1)^{r+1} \frac{(2r)!\zeta(2r)}{r(1-4^{-r})(2\pi)^{2r}}, \quad \kappa_{2r+1}(Y) = 0. \quad (55)$$

Hence

$$\mathbb{E}Y^{2n} = B_{2n}(0, \kappa_2, 0, \kappa_4, \dots, 0, \kappa_{2n}), \quad (56)$$

where the B_{2n} on the right is the complete exponential Bell polynomial, not a Bernoulli number. The first values are

$$\mathbb{E}Y^2 = \frac{1}{9}, \quad \mathbb{E}Y^4 = \frac{19}{675}, \quad \mathbb{E}Y^6 = \frac{583}{59535}. \quad (57)$$

Moments of $X = (1 + Y)/2$ follow by binomial expansion.

6.4 Wick rotation, Newton recurrence, and q -MZV formula

Define

$$\mathcal{A}(w) = \Phi(i\sqrt{w}) = \prod_{n \geq 1} \left(1 + \frac{w}{n^2}\right)^{a_n} = \sum_{r \geq 0} C_r w^r. \quad (58)$$

Then

$$C_r = \frac{(2\pi)^{2r}}{(2r)!} \mathbb{E}Y^{2r}. \quad (59)$$

Writing

$$p_r = \sum_{n \geq 1} \frac{a_n}{n^{2r}} = \frac{\zeta(2r)}{1-4^{-r}}, \quad (60)$$

Newton's recurrence is

$$rC_r = \sum_{j=1}^r (-1)^{j+1} p_j C_{r-j}, \quad C_0 = 1. \quad (61)$$

At $q = 1/4$, the active frontier report further expands C_r by partitions. If $\lambda = (\lambda_1, \dots, \lambda_\ell) \vdash r$, $m_j(\lambda)$ is the multiplicity of part j , and

$$Z_\lambda = \sum_{\substack{n_1, \dots, n_\ell \geq 1 \\ n_i \neq n_j}} \frac{1}{n_1^{2\lambda_1} \dots n_\ell^{2\lambda_\ell}}, \quad (62)$$

then

$$C_r = \sum_{\lambda \vdash r} \frac{Z_\lambda}{\prod_{j \geq 1} m_j(\lambda)!} \prod_{i=1}^{\ell(\lambda)} \frac{q^{\lambda_i(\lambda_i-1)/2}}{(q; q)_{\lambda_i}}, \quad q = \frac{1}{4}. \quad (63)$$

Moreover Z_λ is a symmetrized multiple-zeta value. This formula keeps both the integer spectrum and its dyadic layer occupancy visible.

6.5 Appell and logistic representations already in the corpus

The Appell sequence associated with the compact Rvachev law is defined by

$$\frac{e^{xt}}{M_Y(t)} = \sum_{n \geq 0} A_n(x) \frac{t^n}{n!}. \quad (64)$$

The audited Barnes/inverse draft introduces independent standard logistic variables Λ_j and

$$Z = \sum_{j \geq 1} \frac{\Lambda_j}{\pi 2^j}, \quad (65)$$

with

$$\mathbb{E} e^{itZ} = \frac{1}{M_Y(t)}, \quad A_n(x) = \mathbb{E}(x + iZ)^n. \quad (66)$$

This logistic dual is baseline. The new result in [Section 10](#) is its exact identification as a Gaussian variance mixture driven by the independently developed Thorin variable.

7 Thue–Morse, dyadic arithmetic, and finite product-to-sum bridges

Let $\varepsilon_k = (-1)^{s_2(k)}$. The finite Prouhet product is

$$\prod_{j=0}^{N-1} (1 - z^{2^j}) = \sum_{k=0}^{2^N-1} \varepsilon_k z^k. \quad (67)$$

With $z = e^t$ this becomes

$$\sum_{k < 2^N} \varepsilon_k e^{kt} = \prod_{j=0}^{N-1} (1 - e^{2^j t}). \quad (68)$$

Coefficient comparison yields

$$\sum_{k < 2^N} \varepsilon_k k^m = 0 \quad (0 \leq m < N), \quad \sum_{k < 2^N} \varepsilon_k k^N = (-1)^N N! 2^{N(N-1)/2}. \quad (69)$$

These cancellations explain the compactness of the truncated-power spline (19); they also power the exact dyadic-value formulae, Gaussian-binomial interpolation, and Richardson extrapolants in the repository.

The finite Fourier product

$$\Phi_N(z) = \prod_{h=0}^{N-1} \operatorname{sinc} \left(\frac{\pi z}{2^h} \right) \quad (70)$$

can be converted into a signed exponential sum by expanding the sine factors. Conversely, Prouhet cancellations force high-order zeros in those finite sums. This is the basic product-to-sum bridge connecting Fourier analysis to exact binary arithmetic.

8 Endpoint Laplace phases, Lambert W , and inverse asymptotics

The active frontier synthesis writes

$$g(s) = \log L_X(s) = -\frac{(\log s)^2}{2 \log 2} + \frac{1}{2} \log s + R(\log_2 s) + E(s), \quad (71)$$

where R is real-analytic and one-periodic and

$$0 \leq E(s) \leq \frac{e^{-s}}{(1 - e^{-s})^2}. \quad (72)$$

The CDF integral (51) has saddle equation

$$x = -g'(s), \quad (73)$$

whose exact phase is solved by a lower branch of Lambert W , perturbed by the periodic function and its derivatives. The periodic term is essential: every nonzero Fourier mode is explicitly nonzero in the formal exposition, so replacing it by a constant loses the claimed sharp error scale.

The inverse $Q = F^{-1}$ consequently lives on the stretched-logarithmic scale

$$Q(y) \asymp \sqrt{\log(1/y)} \exp \left[-\sqrt{2 \log 2 \log(1/y)} \right] \quad (y \downarrow 0), \quad (74)$$

with a sharper periodic Lambert expansion in the audited inverse reports. Reflection gives $Q(1-y) = 1 - Q(y)$. These asymptotic representations complement, rather than replace, the exact Fourier and quantile identities developed below.

New theorem packages

9 The dyadic gamma factorization

9.1 Definition, convergence, and poles

Definition 9.1 (Dyadic gamma function [\[proved here\]](#)). For $z \in \mathbb{C}$, away from the negative integers, define

$$\Gamma_{\text{dy}}(z) = \prod_{h=0}^{\infty} \Gamma(1 + 2^{-h}z). \quad (75)$$

Theorem 9.2 (Meromorphy and Mahler equation [\[proved here\]](#)). The product (75) converges locally uniformly away from its poles and defines a meromorphic function satisfying

$$\Gamma_{\text{dy}}(z) = \Gamma(1 + z)\Gamma_{\text{dy}}(z/2), \quad \Gamma_{\text{dy}}(0) = 1. \quad (76)$$

Its pole at $-n$, $n \geq 1$, has exact order $a_n = 1 + v_2(n)$ and it has no other poles.

Proof. Uniformly on compact sets, $\log \Gamma(1 + w) = -\gamma w + \mathcal{O}(w^2)$ as $w \rightarrow 0$. The sums $\sum_h 2^{-h}$ and $\sum_h 4^{-h}$ converge, so the logarithms form a normally convergent series after finitely many factors are removed. Separating the factor $h = 0$ gives (76). At $z = -n$, the factor indexed by h has a simple gamma pole precisely when $n/2^h$ is a positive integer, namely for $0 \leq h \leq v_2(n)$. Their number is $1 + v_2(n)$. $\square$

9.2 Reflection factorization of the Rvachev transform

Theorem 9.3 (Dyadic gamma factorization [\[proved here\]](#)). For every $z \in \mathbb{C}$,

$$\Phi(z) = \frac{1}{\Gamma_{\text{dy}}(z)\Gamma_{\text{dy}}(-z)}. \quad (77)$$

The equality is entire: at integer z , the poles of the gamma factors produce exactly the zeros of Φ with their arithmetic multiplicities.

Proof. Euler's reflection formula in the shifted form

$$\Gamma(1 + w)\Gamma(1 - w) = \frac{\pi w}{\sin(\pi w)} = \frac{1}{\text{sinc}(\pi w)} \quad (78)$$

gives

$$\frac{1}{\Gamma(1 + 2^{-h}z)\Gamma(1 - 2^{-h}z)} = \text{sinc}(\pi z/2^h).$$

Multiplication over $h \geq 0$ yields the sinc product (24). Normal convergence away from the poles and removable continuation at the integers complete the proof. $\square$

Corollary 9.4 (Gamma integral representations of up and F [\[proved here\]](#)). For real x ,

$$\text{up}(x) = \int_{\mathbb{R}} \frac{e^{2\pi i x \xi}}{\Gamma_{\text{dy}}(\xi)\Gamma_{\text{dy}}(-\xi)} d\xi, \quad (79)$$

and for real x ,

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^{\infty} \frac{\sin(2\pi \xi(2x - 1))}{\xi \Gamma_{\text{dy}}(\xi)\Gamma_{\text{dy}}(-\xi)} d\xi. \quad (80)$$

Both integrals are absolutely convergent after the usual removable interpretation at $\xi = 0$.

9.3 Canonical reciprocal product and power series

Let $G_{\text{dy}}(z) = 1/\Gamma_{\text{dy}}(z)$. Euler's canonical product for $1/\Gamma$ and regrouping equal zeros give

$$G_{\text{dy}}(z) = e^{2\gamma z} \prod_{n \geq 1} \left[\left(1 + \frac{z}{n} \right) e^{-z/n} \right]^{a_n}. \quad (81)$$

Consequently

$$\Phi(z) = G_{\text{dy}}(z)G_{\text{dy}}(-z), \quad (82)$$

which splits the even zero divisor into a negative-integer factor and its reflection. For $|z| < 1$,

$$\log \Gamma_{\text{dy}}(z) = -2\gamma z + \sum_{m \geq 2} \frac{(-1)^m \zeta(m)}{m(1 - 2^{-m})} z^m. \quad (83)$$

The even part of (83) is exactly $-\frac{1}{2} \log \Phi(z)$, while the odd part is new information lost in the even Fourier product.

9.4 Dyadic digamma and Stieltjes resolvent

Define

$$\Psi_{\text{dy}}(z) = \frac{\Gamma'_{\text{dy}}(z)}{\Gamma_{\text{dy}}(z)}. \quad (84)$$

Termwise differentiation gives

$$\Psi_{\text{dy}}(z) = \sum_{h \geq 0} 2^{-h} \psi(1 + 2^{-h} z), \quad (85)$$

and the Fourier logarithmic derivative becomes

$$\frac{\Phi'(z)}{\Phi(z)} = \Psi_{\text{dy}}(-z) - \Psi_{\text{dy}}(z) = -2z \sum_{n \geq 1} \frac{a_n}{n^2 - z^2}. \quad (86)$$

Thus the arithmetic Stieltjes resolvent is the odd part of a dyadic digamma function. Differentiating repeatedly produces polygamma representations for every logarithmic jet of Φ .

9.5 Malmsten and heat-trace integrals

For $\Re z > -1$, summing Malmsten's integral for $\log \Gamma(1 + z/2^h)$ gives

$$\log \Gamma_{\text{dy}}(z) = \int_0^\infty \left[2ze^{-t} - \frac{\sum_{h \geq 0} (1 - e^{-2^{-h}zt})}{e^t - 1} \right] \frac{dt}{t}. \quad (87)$$

The inner dyadic exponential sum has the entire expansion

$$\sum_{h \geq 0} (1 - e^{-2^{-h}w}) = \sum_{m \geq 1} \frac{(-1)^{m+1} w^m}{m!(1 - 2^{-m})}. \quad (88)$$

A second integral makes the integer spectrum explicit. From (81) and (32),

$$\log \Gamma_{\text{dy}}(z) = -2\gamma z + \int_0^\infty (zt - (1 - e^{-zt})) L_1(t) \frac{dt}{t}, \quad \Re z > -1. \quad (89)$$

This identity is a direct analytic bridge between the gamma factor, Lambert series, and the zero-divisor heat trace.

9.6 Geometric- q extension

Theorem 9.5 (Geometric gamma factorization [\[proved here\]](#)). *Let $0 < q < 1$ and define*

$$\Gamma_{\langle q \rangle}(z) = \prod_{h \geq 0} \Gamma(1 + q^h z), \quad \Phi_q(z) = \prod_{h \geq 0} \text{sinc}(\pi q^h z). \quad (90)$$

Then

$$\Gamma_{\langle q \rangle}(z) = \Gamma(1 + z) \Gamma_{\langle q \rangle}(qz), \quad (91)$$

$$\Phi_q(z) = \frac{1}{\Gamma_{\langle q \rangle}(z) \Gamma_{\langle q \rangle}(-z)}, \quad (92)$$

$$\log \Gamma_{\langle q \rangle}(z) = -\frac{\gamma z}{1 - q} + \sum_{m \geq 2} \frac{(-1)^m \zeta(m)}{m(1 - q^m)} z^m, \quad |z| < 1. \quad (93)$$

For $q = 1/b$ with integer $b \geq 2$, the pole at $-n$ has order $1 + \max\{h : b^h \mid n\}$.

Proof. The convergence argument is identical to the dyadic case, using $\sum q^h$ and $\sum q^{2h}$. Reflection is applied factor by factor, and the power series follows by summing $\sum_h q^{hm} = (1 - q^m)^{-1}$. $\square$

Remark 9.6. $\Gamma_{\langle q \rangle}$ is not the classical Jackson q -gamma function. It is a geometric Mahler product of ordinary gamma functions. Its denominators $1 - q^m$ nevertheless align exactly with the q -series denominators already pervasive in the Fabius–Rvachev theory.

10 Gaussian variance mixtures and the logistic dual

10.1 The existing Thorin variable

The audited frontier corpus already constructs, for $\tau > 0$, a nonnegative random variable V_τ with

$$\mathbb{E} e^{-w V_\tau} = \mathcal{A}(w)^{-\tau}, \quad (94)$$

and grouped gamma representation

$$V_\tau \stackrel{d}{=} \sum_{n \geq 1} H_{n,\tau}, \quad H_{n,\tau} \sim \text{Gamma}(\tau a_n, \text{rate } n^2), \quad (95)$$

with independent summands. Its discrete Thorin measure is

$$U_\tau = \tau \sum_{n \geq 1} a_n \delta_{n^2}. \quad (96)$$

The construction and its complete-Bernstein consequences are [\[repository\]](#). We now connect it to the independently developed logistic dual.

10.2 Equality in law

Theorem 10.1 (Gaussian variance-mixture bridge [\[proved here\]](#)). *Let*

$$T_\tau = \frac{V_\tau}{4\pi^2}, \quad (97)$$

and let $N \sim \mathcal{N}(0, 1)$ be independent of T_τ . Define

$$Z_\tau = \sqrt{2T_\tau} N = \frac{\sqrt{V_\tau}}{\sqrt{2}\pi} N. \quad (98)$$

Then

$$\mathbb{E}e^{itZ_\tau} = \mathcal{A}\left(\frac{t^2}{4\pi^2}\right)^{-\tau}, \quad (99)$$

$$= \prod_{n \geq 1} \left(1 + \frac{t^2}{4\pi^2 n^2}\right)^{-\tau a_n}, \quad (100)$$

$$= \left[\prod_{j \geq 1} \frac{t/2^j}{\sinh(t/2^j)} \right]^\tau, \quad (101)$$

$$= \left[\Gamma_{\text{dy}}\left(\frac{it}{2\pi}\right) \Gamma_{\text{dy}}\left(-\frac{it}{2\pi}\right) \right]^\tau. \quad (102)$$

For $\tau = 1$,

$$\boxed{\sum_{j \geq 1} \frac{\Lambda_j}{\pi 2^j} \stackrel{\text{d}}{=} \frac{\sqrt{V_1}}{\sqrt{2}\pi} N,} \quad (103)$$

where the Λ_j are independent standard logistic variables.

Proof. Conditioning on T_τ gives

$$\mathbb{E}(e^{it\sqrt{2T_\tau}N} \mid T_\tau) = e^{-t^2 T_\tau}.$$

Taking expectation and applying (94) at $w = t^2/(4\pi^2)$ proves (99); (100) follows from (58). Since $\mathcal{A}(t^2/(4\pi^2)) = M_Y(t)$, the reciprocal hyperbolic-sine product (101) follows from (48). Finally, (77) at $z = it/(2\pi)$ gives (102). At $\tau = 1$, the right-hand side is exactly the characteristic function of the repository's logistic series (65); uniqueness of characteristic functions proves (103). $\square$

The bridge is not a restatement of generalized gamma convolution or of the logistic series. It identifies two previously separate noncompact duals of the compact Rvachev law: one built additively from logistic coordinates and the other obtained by subordinating a Gaussian by the arithmetic Thorin variable.

10.3 Convolution semigroup, density, and moments

The family $(Z_\tau)_{\tau>0}$ is a symmetric convolution semigroup:

$$Z_{\tau_1} + Z'_{\tau_2} \stackrel{d}{=} Z_{\tau_1+\tau_2}. \quad (104)$$

Its density is the exact normal scale mixture

$$p_\tau(x) = \mathbb{E} \left[\frac{1}{\sqrt{4\pi T_\tau}} \exp \left(-\frac{x^2}{4T_\tau} \right) \right]. \quad (105)$$

The cumulants are

$$\kappa_{2r}(Z_\tau) = \tau \frac{(2r)!}{r(4\pi^2)^r} \frac{\zeta(2r)}{1-4^{-r}}, \quad \kappa_{2r+1}(Z_\tau) = 0. \quad (106)$$

In particular, $\text{Var } Z_\tau = \tau/9$. Conditional Gaussian moments yield

$$\mathbb{E} Z_\tau^{2m} = 2^m (2m-1)!! \mathbb{E} T_\tau^m, \quad (107)$$

where

$$\kappa_r(T_\tau) = \tau(r-1)! \frac{\zeta(2r)}{(4\pi^2)^r (1-4^{-r})}. \quad (108)$$

Thus (107) and complete Bell polynomials give a second moment calculus for the logistic/Appell dual.

10.4 Closed Levy density and self-decomposability

Theorem 10.2 (Levy density of the dual semigroup [\[proved here\]](#)). *The characteristic exponent of Z_τ is*

$$-\log \mathbb{E} e^{itZ_\tau} = 2\tau \int_0^\infty (1 - \cos(tx)) L_1(2\pi x) \frac{dx}{x}. \quad (109)$$

Consequently its symmetric Levy measure has density

$$\nu_\tau(x) = \frac{\tau}{|x|} L_1(2\pi|x|) = \frac{\tau}{|x|} \sum_{h \geq 0} \frac{1}{e^{2^{h+1}\pi|x|} - 1}, \quad x \neq 0. \quad (110)$$

Every Z_τ is self-decomposable.

Proof. Use

$$\log \left(1 + \frac{t^2}{\lambda^2} \right) = 2 \int_0^\infty (1 - \cos(tx)) e^{-\lambda x} \frac{dx}{x}$$

in (100), then sum over n . This proves (109) and (110). A symmetric infinitely divisible law with Levy density $k(|x|)/|x|$ is self-decomposable when k is decreasing. Here $k(x) = \tau L_1(2\pi x)$ is a positive sum of decreasing exponentials. $\square$

10.5 Complete monotonicity in the squared coordinate

Corollary 10.3 (Radial positivity in every dimension [\[proved here\]](#)). *For every $\tau > 0$, the function*

$$g_\tau(s) = p_\tau(\sqrt{s}), \quad s > 0, \quad (111)$$

is completely monotone:

$$(-1)^m g_\tau^{(m)}(s) \geq 0 \quad (m \geq 0, s > 0). \quad (112)$$

Consequently $x \mapsto p_\tau(\|x\|)$ is positive definite on $\mathbb{R}^d$ for every $d \geq 1$.

Proof. Formula (105) writes

$$g_\tau(s) = \mathbb{E} \left[(4\pi T_\tau)^{-1/2} e^{-s/(4T_\tau)} \right],$$

a positive mixture of exponentials in s . Bernstein's theorem gives complete monotonicity, and Schoenberg's theorem converts complete monotonicity of g_τ into radial positive definiteness of $g_\tau(\|x\|^2) = p_\tau(\|x\|)$ in every dimension. $\square$

11 Jensen means, factorial products, and binary digit sums

11.1 Exact circular mean

Define the Jensen mean

$$J(r) = \frac{1}{2\pi} \int_0^{2\pi} \log |\Phi(re^{i\theta})| \, d\theta, \quad r > 0, \quad (113)$$

with the improper interpretation when the circle passes through an integer zero.

Theorem 11.1 (Arithmetic Jensen formula [\[proved here\]](#)). *Let $N = \lfloor r \rfloor$ and $N_j = \lfloor N/2^j \rfloor$. Then*

$$J(r) = 2 \sum_{n \leq N} a_n \log \frac{r}{n}, \quad (114)$$

$$= 2 \sum_{j \geq 0} \left[N_j \log \frac{r}{2^j} - \log \Gamma(N_j + 1) \right]. \quad (115)$$

The second sum is finite. Equivalently,

$$\exp \left(\frac{J(r)}{2} \right) = \prod_{n \leq N} \left(\frac{r}{n} \right)^{a_n} = \prod_{j \geq 0} \frac{(r/2^j)^{N_j}}{N_j!}. \quad (116)$$

Proof. Jensen's formula for the entire function Φ , normalized by $\Phi(0) = 1$, says that the mean is the sum of $\log(r/|\rho|)$ over zeros ρ in the disk, counted with multiplicity. The zeros occur at $\pm n$ with multiplicity a_n , giving (114). Expand $a_n = \sum_{j \geq 0} \mathbf{1}_{2^j | n}$, write $n = 2^j m$, and sum $\log(r/(2^j m))$ for $1 \leq m \leq N_j$. This proves (115) and (116). $\square$

11.2 Digit-sum derivative

Corollary 11.2 (Binary slope law [\[proved here\]](#)). *For $N < r < N + 1$,*

$$\boxed{J'(r) = \frac{2A(N)}{r} = \frac{2(2N - s_2(N))}{r}.} \quad (117)$$

Thus

$$\frac{rJ'(r)}{2} = 2 \lfloor r \rfloor - s_2(\lfloor r \rfloor) \quad (118)$$

recovers the binary digit sum directly from a circular mean of the Fourier image.

Remark 11.3. The zero-counting law $A(N) = 2N - s_2(N)$ was already in the repository. The new point is that it is the logarithmic radial derivative of a two-dimensional complex-analytic average, and that the full mean has the factorial product (116).

11.3 Asymptotic growth

Theorem 11.4 (Leading Jensen asymptotic [\[proved here\]](#)). *As $r \rightarrow \infty$,*

$$\boxed{J(r) = 4r - \frac{(\log r)^2}{2 \log 2} + \mathcal{O}(\log r).} \quad (119)$$

Proof. Let $N = \lfloor r \rfloor$. Partial summation applied to (114) gives

$$\frac{J(r)}{2} = A(N) \log(r/N) + \int_1^N \frac{A(\lfloor t \rfloor)}{t} dt.$$

The first term is $\mathcal{O}(1)$. Since $A(n) = 2n - s_2(n)$, the linear part contributes $2N + \mathcal{O}(\log N)$. The elementary summatory digit estimate

$$\sum_{n \leq x} s_2(n) = \frac{x \log x}{2 \log 2} + \mathcal{O}(x)$$

and another partial summation give

$$\int_1^N \frac{s_2(\lfloor t \rfloor)}{t} dt = \frac{(\log N)^2}{4 \log 2} + \mathcal{O}(\log N).$$

Multiply by two and replace N with r . □

11.4 Subunit circular energy

For $0 \leq r < 1$, Jensen's mean vanishes because the disk contains no zeros. The angular fluctuation nevertheless has an exact zeta-series energy. Put

$$c_m = \frac{\zeta(2m)}{m(1 - 4^{-m})}. \quad (120)$$

Then (28) and trigonometric orthogonality give

$$\boxed{\frac{1}{2\pi} \int_0^{2\pi} \left(\log |\Phi(re^{i\theta})| \right)^2 d\theta = \frac{1}{2} \sum_{m \geq 1} c_m^2 r^{4m}.} \quad (121)$$

More generally, for $0 \leq r, s < 1$,

$$\frac{1}{2\pi} \int_0^{2\pi} \log |\Phi(re^{i\theta})| \log |\Phi(se^{i\theta})| d\theta = \frac{1}{2} \sum_{m \geq 1} c_m^2 (rs)^{2m}. \quad (122)$$

This is a local angular covariance kernel for the Fourier determinant.

12 Non-holonomicity from the arithmetic zero divisor

Theorem 12.1 (The Rvachev transform is not D-finite [\[proved here\]](#)). *The entire function Φ does not satisfy any nontrivial linear differential equation*

$$p_R(z)y^{(R)} + p_{R-1}(z)y^{(R-1)} + \cdots + p_0(z)y = 0 \quad (123)$$

with polynomial coefficients $p_j \in \mathbb{C}[z]$. Equivalently, Φ is not holonomic over $\mathbb{C}(z)$.

Proof. Suppose (123) holds with $p_R \neq 0$. At any ordinary point z_0 with $p_R(z_0) \neq 0$, uniqueness for the associated analytic initial-value problem implies that a nonzero solution cannot have a zero of multiplicity R or larger: if the first R derivatives vanish, the solution is identically zero near z_0 .

The polynomial p_R has only finitely many zeros. But Φ has a zero at 2^k of multiplicity $k + 1$. Choose $k \geq R$ such that $p_R(2^k) \neq 0$. This gives an ordinary zero of multiplicity at least R , a contradiction. $\square$

Corollary 12.2 (Coefficient and algebraic consequences [\[proved here\]](#)). *The Maclaurin coefficient sequence of Φ is not P-recursive. In particular, Φ is not algebraic over $\mathbb{C}(z)$.*

Proof. A power series is D-finite if and only if its coefficient sequence is P-recursive. Every algebraic power series is D-finite. $\square$

Proposition 12.3 (The dyadic gamma factors are not D-finite [\[proved here\]](#)). *Neither Γ_{dy} nor $G_{\text{dy}} = 1/\Gamma_{\text{dy}}$ is D-finite.*

Proof. A D-finite germ has only finitely many possible finite singularities, whereas Γ_{dy} has poles at every negative integer. The reciprocal G_{dy} is entire but has a zero at -2^k of multiplicity $k + 1$, so the preceding multiplicity argument applies. $\square$

Remark 12.4. Non-D-finiteness is much weaker than differential transcendence. The latter is stated only as a conjecture in [Section 17](#).

13 Fourier representations of the bounded Fabius function

13.1 The density transform

Let $f_X = F' = 2 \text{up}(2x - 1)$ on $[0, 1]$ and zero outside. A change of variables gives

$$\widehat{f_X}(\xi) = e^{-\pi i \xi} \Phi(\xi/2). \quad (124)$$

Hence the compactly supported pieces

$$F_0(x) = F(x)\mathbf{1}_{[0,1]}(x), \quad S_0(x) = (1 - F(x))\mathbf{1}_{[0,1]}(x) \quad (125)$$

have, for $\xi \neq 0$,

$$\widehat{F}_0(\xi) = \frac{e^{-\pi i \xi} \Phi(\xi/2) - e^{-2\pi i \xi}}{2\pi i \xi}, \quad (126)$$

$$\widehat{S}_0(\xi) = \frac{1 - e^{-\pi i \xi} \Phi(\xi/2)}{2\pi i \xi}. \quad (127)$$

Both values at zero are $1/2$. Inserting (24), (26), or (77) turns these into explicit product representations of the Fourier image of the bounded CDF.

13.2 The global CDF as a tempered distribution

Let H be the Heaviside distribution. Since $F = H * f_X$,

$$\widehat{H} = \frac{1}{2} \delta_0 + \frac{1}{2\pi i} \text{pv} \frac{1}{\xi} \quad (128)$$

gives the distributional formula

$$\boxed{\widehat{F} = \frac{1}{2} \delta_0 + \frac{1}{2\pi i} \text{pv} \left(\frac{e^{-\pi i \xi} \Phi(\xi/2)}{\xi} \right)}. \quad (129)$$

This is the correct Fourier image of the globally bounded, nonintegrable CDF. Omitting the Dirac mass or principal value changes the distribution.

13.3 Sine integral and odd-harmonic series

Since $F(x) = \frac{1}{2} + \int_0^{2x-1} \text{up}(y) dy$, Fourier inversion yields

$$\boxed{F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Phi(\xi) \frac{\sin(2\pi \xi (2x-1))}{\xi} d\xi}. \quad (130)$$

This holds for all real x under the bounded-CDF convention. Integrating the interval cosine series (44) gives a second representation.

Theorem 13.1 (Odd-harmonic Fabius series [\[proved here\]](#)). *For $0 \leq x \leq 1$,*

$$F'(x) = 1 - 2 \sum_{k \geq 0} \Phi\left(k + \frac{1}{2}\right) \cos(2\pi(2k+1)x), \quad (131)$$

$$\boxed{F(x) = x - \frac{1}{\pi} \sum_{k \geq 0} \frac{\Phi(k+1/2)}{2k+1} \sin(2\pi(2k+1)x)}. \quad (132)$$

Both series converge absolutely and uniformly together with every differentiated series allowed by the displayed factors.

Proof. Substitute $t = 2x - 1$ into (44) and multiply by two. Since

$$\cos((2k+1)\pi(2x-1)) = -\cos(2\pi(2k+1)x),$$

we obtain (131). Integrating from 0 to x and using $F(0) = 0$ gives (132). Rapid decay of the half-integer samples follows from $\text{up} \in C_c^\infty$. $\square$

13.4 Gamma versions

The new factorization makes the preceding formulae explicitly gamma-theoretic:

$$\widehat{f_X}(\xi) = \frac{e^{-\pi i \xi}}{\Gamma_{\text{dy}}(\xi/2)\Gamma_{\text{dy}}(-\xi/2)}, \quad (133)$$

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \frac{\sin(2\pi\xi(2x-1))}{\xi\Gamma_{\text{dy}}(\xi)\Gamma_{\text{dy}}(-\xi)} d\xi. \quad (134)$$

Thus ordinary gamma functions, assembled on a dyadic scale, provide direct integral and Fourier images for the Fabius CDF.

14 Quantile and inverse-Fabius transform representations

14.1 Quantile pushforward and product transforms

Let $Q = F^{-1}$ be the clamped inverse. The current formal exposition proves the support-free quantile substitution theorem

$$\int_0^1 \varphi(Q(y)) dy = \mathbb{E}\varphi(X) \quad (135)$$

for a broad Banach-valued class of integrands. Therefore

$$\int_0^1 e^{-sQ(y)} dy = \prod_{j \geq 1} \frac{1 - e^{-s/2^j}}{s/2^j}, \quad (136)$$

$$\int_0^1 e^{s(Q(y)-1/2)} dy = \prod_{j \geq 1} \frac{\sinh(s/2^{j+1})}{s/2^{j+1}}, \quad (137)$$

$$\int_0^1 e^{-2\pi i \xi Q(y)} dy = e^{-\pi i \xi} \Phi(\xi/2). \quad (138)$$

For $0 < \Re \alpha < 1$,

$$\int_0^1 Q(y)^\alpha dy = \frac{\alpha}{\Gamma(1-\alpha)} \int_0^\infty \left[1 - \prod_{j \geq 1} \frac{1 - e^{-t/2^j}}{t/2^j} \right] t^{-\alpha-1} dt. \quad (139)$$

These are [\[derived synthesis\]](#)consequences of the now-formal quantile theorem.

14.2 A nonlinear Fourier transform of the inverse

Theorem 14.1 (Inverse-transform identity [\[proved here\]](#)). *For $\eta \in \mathbb{C} \setminus \{0\}$,*

$$\int_0^1 Q(y) e^{-2\pi i \eta y} dy = \int_0^1 x F'(x) e^{-2\pi i \eta F(x)} dx, \quad (140)$$

$$= \frac{\int_0^1 e^{-2\pi i \eta F(x)} dx - e^{-2\pi i \eta}}{2\pi i \eta}. \quad (141)$$

The value at $\eta = 0$ is $1/2$ by continuity.

Proof. Apply $y = F(x)$, $dy = F'(x) dx$, and $Q(F(x)) = x$ to obtain (140). Since $de^{-2\pi i \eta F(x)} = -2\pi i \eta F'(x) e^{-2\pi i \eta F(x)} dx$, integration by parts gives (141). $\square$

14.3 Sine series of the inverse in terms of F

Theorem 14.2 (Nonlinear inverse-Fabius sine series [\[proved here\]](#)). *Put*

$$c_n = \frac{1}{\pi n} \int_0^1 \cos(2\pi n F(x)) \, dx, \quad n \geq 1. \quad (142)$$

Then

$$Q(y) = y + \sum_{n \geq 1} c_n \sin(2\pi n y), \quad 0 \leq y \leq 1. \quad (143)$$

The series converges in $L^2(0, 1)$ and pointwise at every $y \in [0, 1]$. Moreover $c_n = o(1/n)$.

Proof. The function $g(y) = Q(y) - y$ is continuous, vanishes at both endpoints, and is of bounded variation. The symmetry $Q(1 - y) = 1 - Q(y)$ gives $g(1 - y) = -g(y)$, so only the modes $\sin(2\pi n y)$ occur. Their coefficients are

$$2 \int_0^1 g(y) \sin(2\pi n y) \, dy.$$

For the Q term, use $y = F(x)$ and integrate by parts:

$$\begin{aligned} \int_0^1 Q(y) \sin(2\pi n y) \, dy &= \int_0^1 x F'(x) \sin(2\pi n F(x)) \, dx \\ &= -\frac{1}{2\pi n} + \frac{1}{2\pi n} \int_0^1 \cos(2\pi n F(x)) \, dx. \end{aligned}$$

Since $\int_0^1 y \sin(2\pi n y) \, dy = -1/(2\pi n)$, the difference is exactly (142). Dirichlet–Jordan convergence applies to the bounded-variation function g . Finally,

$$\int_0^1 \cos(2\pi n F(x)) \, dx = \int_0^1 \cos(2\pi n y) Q'(y) \, dy$$

with $Q' \in L^1(0, 1)$; the Riemann–Lebesgue lemma gives $c_n = o(1/n)$. $\square$

Remark 14.3. Formula (143) is not an asymptotic reversion. It is a global interval representation of the inverse in which every coefficient is a nonlinear oscillatory integral of the original CDF. It therefore complements the repository’s local Lambert–Bell inverse expansions.

15 Compact representation atlas

The following table is a lookup index rather than a proof substitute. It collects the main representations developed above and the principal established forms in the audited corpus. The domain column records the natural interpretation: pointwise, entire, distributional, probabilistic, or asymptotic.

Object	Representation	Domain	Status
F	$F(x) = \mathbb{P}(\sum_{j \geq 1} 2^{-j} V_j \leq x)$	all $x \in \mathbb{R}$	[repo]
up	$\mathbb{P}(Y \in dy) = \text{up}(y) \, dy, Y = \sum 2^{-j} U_j$	probability	[repo]
F, up	$F'(x) = 2 \text{up}(2x - 1)$	$0 < x < 1$	[repo]
up	$\text{up}(x) = \int_{2x-1}^{2x+1} \text{up}(t) \, dt$	pointwise	[repo]
up'	$\text{up}'(x) = 2 \text{up}(2x + 1) - 2 \text{up}(2x - 1)$	pointwise	[repo]

Object	Representation	Domain	Status
F	fixed integral equation (8)	$[0, 1]$	[repo]
$\mathcal{F}^{(m)}$	$2^{\binom{m+1}{2}} \mathcal{F}(2^m x)$	all orders	[repo]
up	$*_{j \geq 1} (2^{j-1} \mathbf{1}_{[-2^{-j}, 2^{-j}]})$	L^1 /probability	[repo]
u_N	finite box convolution (14)	compact spline	[repo]
u_N	Thue–Morse truncated powers (19)	pointwise	[repo]
up	$u_N * (2^N \text{up}(2^N \cdot))$	exact head–tail	[repo]
F	expectation/cube recursion (17)	exact	[repo]
Φ	$\prod_{h \geq 0} \text{sinc}(\pi z / 2^h)$	entire	[repo]
Φ	$\prod_{m \geq 1} (\cos(\pi z / 2^m))^m$	entire	[repo]
Φ	$\prod_{n \geq 1} (1 - z^2 / n^2)^{1+v_2(n)}$	entire	[repo]
Φ	$[\Gamma_{\text{dy}}(z) \Gamma_{\text{dy}}(-z)]^{-1}$	entire	[new]
Φ	$\det(I - z^2 D^{-2})$	Fredholm	[repo]
$\log \Phi$	zeta series (28)	$ z < 1$	[repo]
Φ' / Φ	arithmetic resolvent (29)	off zeros	[repo]
$\log \Phi$	Frullani–Lambert integral (36)	$ \Re z < 1$	[derived]
up	Fourier inversion (40)	all real x	[repo]
up	cosine integral (41)	all real x	[repo]
up	half-integer cosine series (44)	$[-1, 1]$	[repo]
Φ	Shannon sampling (46)	all $z \in \mathbb{C}$	[repo]
up	$\sum_{k \in \mathbb{Z}} \text{up}(x - k) = 1$	all x	[repo]
up	Legendre series (47)	$[-1, 1]$	[repo]
M_Y	$\prod_{j \geq 1} \sinh(t/2^j)/(t/2^j)$	entire	[repo]
M_X	$\prod_{j \geq 1} (e^{t/2^j} - 1)/(t/2^j)$	entire	[repo]
L_X	$\prod_{j \geq 1} (1 - e^{-s/2^j})/(s/2^j)$	entire in s	[repo]
F	Bromwich integral (51)	$c > 0$	[repo]
$\mathbb{E}X^{-\alpha}$	Mellin–Laplace integral (52)	$\Re \alpha > 0$	[derived]
$\mathbb{E}X^\alpha$	subtracted Mellin integral (53)	$0 < \Re \alpha < 1$	[derived]
$\mathbb{E}Y^{2n}$	Bernoulli cumulants and Bell polynomial (56)	exact	[repo]
C_r	Newton recurrence (61)	exact	[repo]
C_r	q -binomial/MZV partition sum (63)	$q = 1/4$	[repo]
F	density transform (124)	entire in ξ	[new]
F_0	compact transform (126)	$\xi \in \mathbb{R}$	[new]
F	tempered transform (129)	$\mathcal{S}'$	[new]
F	sine integral (130)	all real x	[new]
F	odd-harmonic sine series (132)	$[0, 1]$	[new]
Γ_{dy}	gamma product (75)	meromorphic	[new]
$1/\Gamma_{\text{dy}}$	arithmetic canonical product (81)	entire	[new]
Γ_{dy}	heat-trace integral (89)	$\Re z > -1$	[new]
V_τ	gamma convolution (95)	probability	[repo]
Z	dyadic logistic sum (65)	probability	[repo]
Z_τ	Gaussian variance mixture (98)	probability	[new]
Z_τ	gamma/sinc characteristic products (101)–(102)	real t	[new]
Z_τ	normal-mixture density (105)	real x	[new]
Z_τ	Levy density (110)	$x \neq 0$	[new]

Object	Representation	Domain	Status
$J(r)$	zero sum and factorial product (114)–(116)	$r > 0$	[new]
$J'(r)$	$(4 \lfloor r \rfloor - 2s_2(\lfloor r \rfloor))/r$	noninteger r	[new]
$J(r)$	$4r - (\log r)^2/(2 \log 2) + \mathcal{O}(\log r)$	$r \rightarrow \infty$	[new]
Q	quantile pushforward (135)	measurable tests	[repo]
Q	Laplace/product transform (136)	complex s	[derived]
Q	nonlinear Fourier transform (141)	$\eta \neq 0$	[new]
Q	sine series (143)	$[0, 1]$	[new]
F, Q	periodic lower-Lambert endpoint expansions	endpoint	[repo]

16 Numerical experiments and reproducibility

The package contains `numerical_experiments.py`, written to be run without network access. It uses double precision, a fixed seed 20260827, vectorized dyadic products, and explicit tail diagnostics. The program creates the figures in this section and writes a plain-text audit trail to `figures/numerical_results.txt`.

16.1 Dyadic gamma factorization

Three complex test points were evaluated with sixty product layers. The absolute errors in (77) were between 1.8×10^{-15} and 3.6×10^{-15} ; the Mahler functional equation (76) was verified at comparable accuracy. The finite-layer identity is exact mathematically, so the plateau in Fig. 1 is floating-point accumulation rather than truncation error.

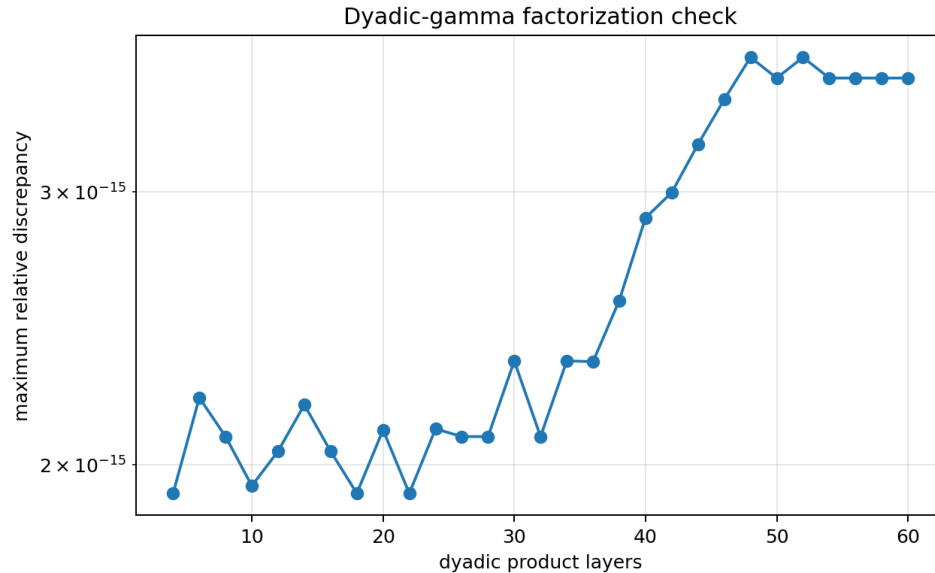


Figure 1: Finite-layer check of the dyadic-gamma reflection factorization.

16.2 Jensen mean

Angular trapezoidal quadrature with 8192 nodes was compared with the exact zero-divisor formula on 76 radii. The maximum absolute discrepancy was 7.11×10^{-15} . Representative values are

r	angular quadrature	exact formula	error
0.6	-3.6082×10^{-16}	0	-3.61×10^{-16}
1.4	0.6729444732424255	0.6729444732424258	-3.33×10^{-16}
3.7	5.496849257625430	5.496849257625430	0
10.3	26.15154490220755	26.15154490220754	3.55×10^{-15}

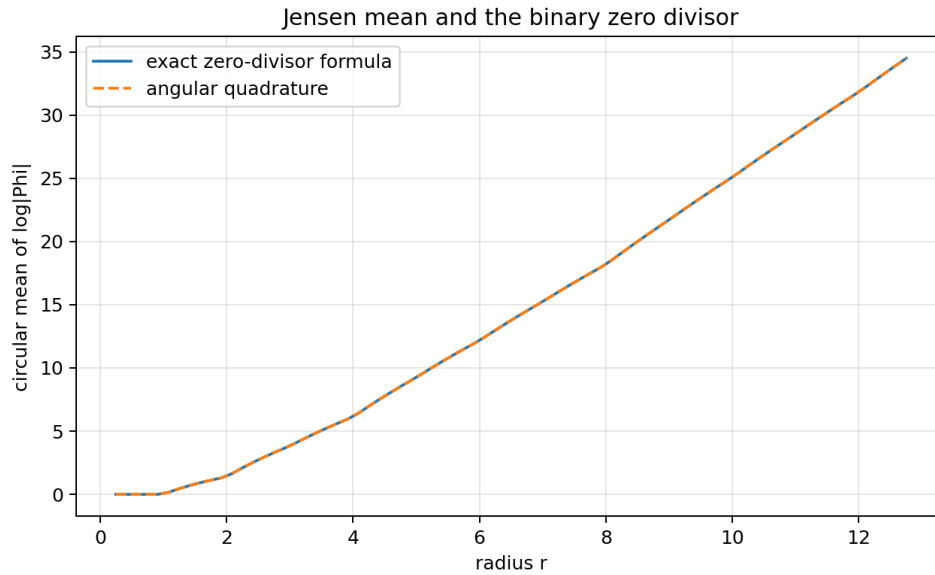


Figure 2: Jensen’s angular mean versus the exact factorial/zero-divisor expression.

16.3 Logistic series versus Gaussian mixture

The simulation used 100,000 samples. The logistic series retained sixteen dyadic layers; its omitted standard deviation was 5.09×10^{-6} . The gamma mixture retained the first 250 spectral sites and added the exact mean of the positive tail. The tail standard deviation was 2.07×10^{-4} , negligible compared with the exact standard deviation $1/3$ of Z_1 . The two-sample Kolmogorov–Smirnov statistic was 0.00347 with p-value 0.5825. Sample variances were 0.1114725 and 0.1110627, compared with the exact $1/9 = 0.111111\dots$

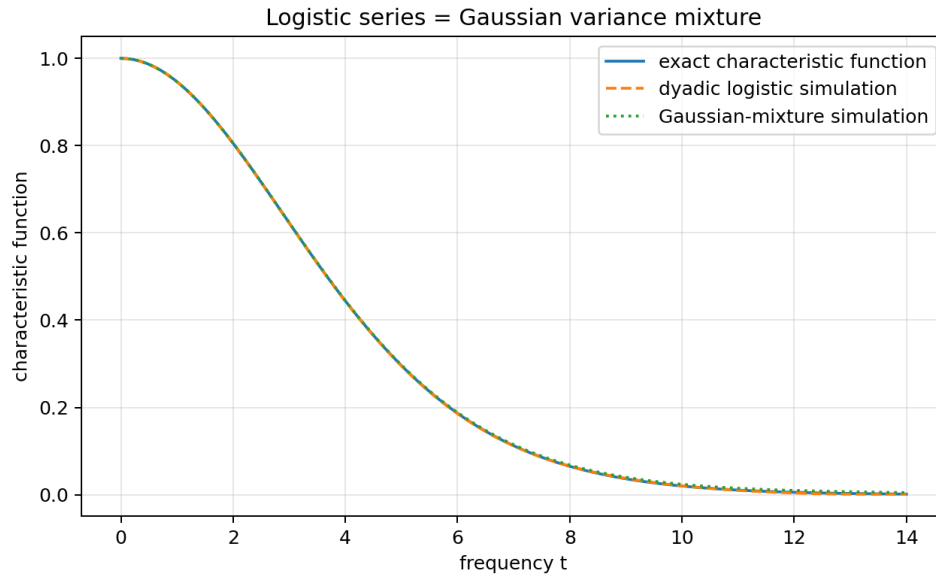


Figure 3: The exact characteristic function and two independent simulations of the law in (103).

16.4 Fabius sine series

The series (132) was compared with the independent Fourier integral (130). At $x = 0.25$, for example,

$$F(0.25) = 0.06944444445136150 \quad (\text{integral}), \quad 0.06944444444444434 \quad (128 \text{ harmonics}).$$

The discrepancy is governed by the numerical quadrature floor after roughly thirty-two harmonics. The rapid early convergence is visible in Fig. 4.

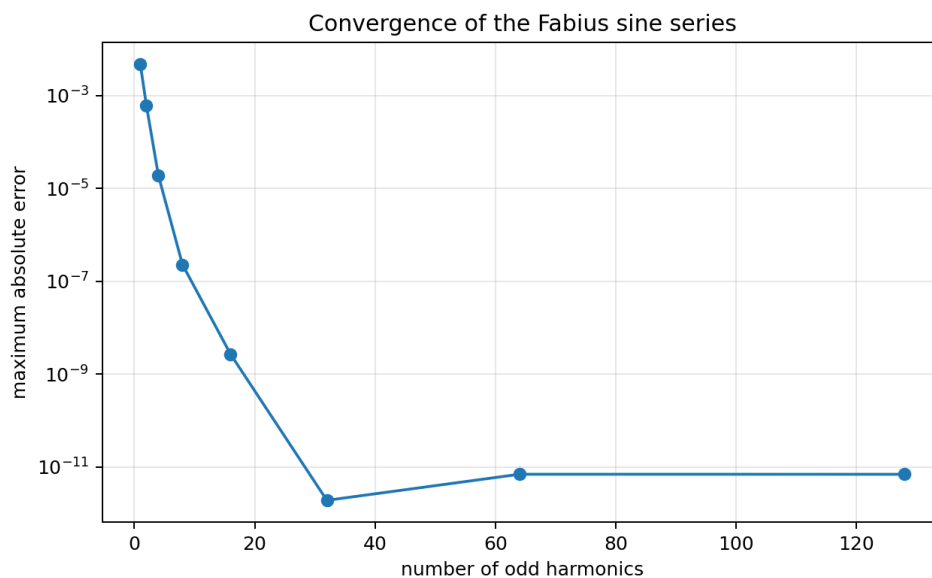


Figure 4: Maximum error of the odd-harmonic Fabius series against Fourier inversion on ten sample points.

16.5 Reproduction command

From the extracted package directory, run

```
python numerical_experiments.py
```

The script creates no hidden cache and writes all outputs beneath **figures/**. The source comments explain the product truncations, gamma-tail correction, and quadrature splitting at integer zeros.

17 Conjectures and testable frontier statements

Conjecture 17.1 (Differential transcendence of the Fourier determinant **[conjecture]**). *The functions Φ , G_{dy} , and Γ_{dy} are differentially transcendental over $\mathbb{C}(z)$: none satisfies a nonzero algebraic differential equation with coefficients in $\mathbb{C}(z)$.*

The non-D-finite theorem is evidence but does not approach the full differential-algebraic claim. A promising route is to combine the Mahler equation (76) with differential Galois theory for mixed differential–difference systems and the nonperiodic pole multiplicities.

Conjecture 17.2 (Full Delange expansion of the Jensen mean **[conjecture]**). *There exist explicit continuous one-periodic functions P_1, P_0 such that*

$$J(r) = 4r - \frac{(\log r)^2}{2 \log 2} + (\log r)P_1(\log_2 r) + P_0(\log_2 r) + o(1), \quad (144)$$

away from a controlled boundary layer around the integers. The functions should be computable from the Delange Fourier series for the binary digit sum and from the fractional part $r - \lfloor r \rfloor$.

Conjecture 17.3 (Asymptotics of inverse Fourier coefficients **[conjecture]**). *Let c_n be defined by (142). Then there is a bounded, nonconstant, phase-dependent function P such that*

$$\pi n c_n = Q(1/n) (P(\log_2 \lambda_n) + o(1)), \quad (145)$$

where λ_n is the exact lower-Lambert saddle phase for $F(1/n)$. In particular, $\log |n c_n|$ should have leading scale $-\sqrt{2 \log 2 \log n}$ with the same periodic modulation that governs the inverse endpoint.

This statement is testable by computing the oscillatory integral in (142) with the repository’s certified endpoint saddle machinery. It would turn the local inverse asymptotic into a global spectral law for Q .

Conjecture 17.4 (Hyperbolic complete monotonicity of the mixture density **[conjecture]**). *For every $\tau > 0$, the function $s \mapsto p_\tau(\sqrt{s})$ is hyperbolically completely monotone, not merely completely monotone. Equivalently, the normal variance mixture may lie in the Bondesson class with a stronger multiplicative total-positivity structure inherited from the discrete Thorin measure.*

Conjecture 17.5 (Arithmetic special values of the dyadic gamma function **[conjecture]**). *At rational arguments with odd denominator, suitable products of $\Gamma_{\text{dy}}(r)$, powers of 2π , and classical Barnes multiple-gamma constants reduce to a finite family of dyadic L -values and logarithmic q -products. The functional equation (76) should produce explicit distribution relations indexed by the binary orbit of r .*

Conjecture 17.6 (Optimal rational approximation from the Stieltjes resolvent **[conjecture]**). *Diagonal Padé approximants obtained from $\Psi_{\text{dy}}(-z) - \Psi_{\text{dy}}(z)$ provide alternating, globally sharp rational bounds for $\Phi(iy)$ and for the characteristic function of Z_τ on $y > 0$, with error constants controlled explicitly by the next unused spectral atom.*

18 Promising research directions

18.1 Formalization roadmap

The shortest Lean route to the new results is modular.

1. Define Γ_{dy} as a normally convergent product of meromorphic gamma factors and prove (76) and (77) by the existing infinite product API.
2. Reuse the formal zero-multiplicity theorem to prove non-D-finiteness in a small analytic-ODE module.
3. Build Z_τ by conditioning a standard Gaussian on the already developed V_τ , then identify characteristic functions.
4. Formalize Jensen’s formula for the canonical product, followed by the finite factorial regrouping. The digit-sum derivative then becomes arithmetic simplification.
5. Derive the compact and distributional Fourier transforms of F from the existing transform of up and a formal Heaviside distribution library.
6. Use the support-free quantile theorem and bounded-variation Fourier convergence to formalize (141) and (143).

18.2 Spectral geometry and complex analysis

The exact Jensen mean suggests treating Φ as an arithmetic spectral determinant in the complex plane, not only on the real axis. Natural next objects include logarithmic capacity, angular large deviations of $\log |\Phi|$, Nevanlinna characteristics refined by binary digit sums, and de Branges spaces generated by the reflected factors $G_{\text{dy}}(z)$ and $G_{\text{dy}}(-z)$.

18.3 Probability and semigroups

The family Z_τ gives a canonical subordinate-Brownian semigroup attached to the Rvachev zero divisor. Its transition density, fractional generator, entropy production, and small- time scaling should be expressible through the two heat kernels L_1 and K . The explicit Levy density (110) makes simulation, self-decomposability, and potential theory accessible without reconstructing the gamma series term by term.

18.4 Base- b and geometric- q families

The geometric gamma factorization (92) should be developed in parallel with the repository’s base-generic Gaussian-binomial interpolation. For integer b , the collision multiplicity of gamma poles is a whole-base valuation; for nonintegral q^{-1} , the pole orbits are typically noncoincident. Comparing these regimes may separate which phenomena are genuinely dyadic from those caused only by geometric scaling.

18.5 Inverse spectral analysis

The coefficients (142) provide a concrete meeting point for global Fourier analysis and endpoint Lambert asymptotics. Certified steepest-descent evaluation of these coefficients could reveal periodic modulation, sign patterns, and optimal approximation rates for the inverse. A parallel study of $\int_0^1 e^{-2\pi i \eta F(x)} dx$ would produce a nonlinear Fourier theory of the CDF itself.

18.6 Computer-assisted certification

The existing Python checks are exploratory. A rigorous package should combine interval arithmetic for gamma products, exact rational moment recurrences, certified quadrature split at the integer zero set, and explicit tail bounds from the rapid Fourier decay. Such a tool could test the conjectural Jensen fluctuation and inverse-coefficient phases at hundreds of dyadic scales without confusing truncation error with genuine periodic oscillation.

19 Conclusion

The Fabius–Rvachev system is unusually rich because the same object can be read as a smooth compact bump, a CDF, an infinite uniform convolution, a Thue–Morse spline limit, an entire sinc determinant, an arithmetic spectrum, a q -series, and a Lambert-saddle problem. The source audit shows that many attractive formulae already coexist in the repository; treating them as separate discoveries would obscure rather than advance the subject.

The new results in this report add bridges rather than isolated identities. The dyadic gamma function splits the even Fourier determinant into reflected arithmetic gamma factors. The Gaussian-mixture theorem joins the logistic/Appell dual to the Thorin/gamma dual and exposes a new Levy semigroup. Jensen’s formula turns binary digit sums into a radial complex-analytic observable. Unbounded multiplicities settle non-holonomicity in one stroke. Finally, the bounded CDF and its inverse acquire transform formulae that are global, explicit, and compatible with the existing endpoint theory. Each bridge supplies both a theorem and a research program, with a comparatively short path to formal verification.

A Audited living document families

The active draft root at the audited snapshot contained the following document families. Their exact names are retained to make future collision searches reproducible.

Draft family	Principal audited themes
Fabius_Arithmetic_Rays_Frontier_Report	arithmetic rays, exact values, spectral sampling
Fabius_Dyadic_Self_Sampling_Frontier_Package	dyadic self-sampling and reconstruction
Fabius_Half_Integer_Spectral_Frontier_Report	half-integer cosine spectrum, Prouhet identities, alias transforms
Fabius_Inverse_Frontier_Report_Source_and_PDF	inverse germs, Catalan and Lambert–Bell asymptotics
Fabius_Newton_Rvachev_Frontier_Report	Newton recurrences, exponent sequences, exact FFT checks
Fabius_Rvachev_Frontier_Report-2	arithmetic spectrum, zero multiplicities, heat trace, lobe signs
Fabius_Rvachev_Frontier_Report	tail quadrature, exact dyadic error laws, q -Richardson acceleration
Fabius_Rvachev_New_Frontiers	q -MZV moments, Thorin dual, complete Bernstein theory, theta completion
Fabius_Rvachev_Thue_Morse_Frontier_Results	finite blocks, deconvolution, zeta–Lambert tails

Draft family	Principal audited themes
Spectral_Arithmetic_Pascal_Rvachev_Hierarchy	spectral arithmetic and Pascal-type hierarchies
Thue_Morse_Formula_Atlas	broad Thue–Morse product, series, automata, and arithmetic atlas
fabius_frontier_dyadic_inverse_barnes_report	dyadic inverse, Barnes prefixes, Appell and logistic dual
fabius_frontier_new_results	logarithmic q -Richardson and Lambert phase locking
fabius_frontier_results	q -Richardson sinc products and acceleration
fabius_frontier_results_bundle	Fredholm determinant, spectral zeta, zero jets, q -Euler determinant
fabius_frontier_spectral_endpoint_report_bundle	endpoint spectral products, Mellin and saddle analysis
rvachev_up_fourier_decay	successive Fourier-decay audits and certified bounds

The source audit also included the primary formal article, the consolidated **non-formalized-research-frontiers.tex**, glossary, Lean walkthrough, non-elementarity document, embedded papers, and the archive provenance map. The archive map was essential for identifying superseded or byte-identical historical variants without treating them as independent sources.

B Numerical package contents

The ZIP archive accompanying this report contains:

- `fabius_rvachev_representation_frontier_report.tex`;
- the compiled PDF with the same basename;
- `numerical_experiments.py`, with detailed comments and deterministic seed;
- `figures/numerical_results.txt`;
- the four PNG figures embedded in the report;
- `SOURCE_AUDIT.md`, a concise audit and novelty-collision record;
- `README.md`, reproduction instructions and file descriptions.

C Selected derivations in one line

For convenient reuse, here are the central bridges in compressed form:

$$\begin{aligned}
\Gamma(1+w)\Gamma(1-w) &= 1/\operatorname{sinc}(\pi w) &\implies & \Phi(z) = 1/[\Gamma_{\mathrm{dy}}(z)\Gamma_{\mathrm{dy}}(-z)], \\
\mathbb{E}e^{-wV_\tau} &= \mathcal{A}(w)^{-\tau}, \quad \mathbb{E}(e^{it\sqrt{2T}N} \mid T) = e^{-t^2T} &\implies & Z_1 \stackrel{\mathrm{d}}{=} \sqrt{2T_1}N, \\
\operatorname{zeros}(\Phi) &= \{\pm n \text{ with multiplicity } a_n\} &\implies & J(r) = 2 \sum_{n \leq r} a_n \log(r/n), \\
a_{2k} &= k+1 \rightarrow \infty &\implies & \Phi \text{ is not D-finite,} \\
F'(x) &= 2 \operatorname{up}(2x-1), \quad \widehat{\operatorname{up}} = \Phi &\implies & \widehat{F}'(\xi) = e^{-\pi i \xi} \Phi(\xi/2), \\
y &= F(x), \quad Q(F(x)) = x &\implies & \widehat{Q}(\eta) = \frac{\int_0^1 e^{-2\pi i \eta F(x)} \mathrm{d}x - e^{-2\pi i \eta}}{2\pi i \eta}.
\end{aligned}$$

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